IILM UNIVERSITY, GREATER NOIDA

INTERNSHIP REPORT PRESENTATION

Google AI-ML

Virtual Internship

AICTE – EduSkills Virtual Internship Program | Supported by Google for Developers

Piyush Kumar Yadav

Enrollment No. 2410030981 | B.Tech CSE — 5th Semester

Organization: Google for Developers | AICTE – EduSkills Program

8 Weeks | April – June 2026

Piyush Kumar Yadav

B.Tech – Computer Science & Engineering

ABOUT THE STUDENT

Candidate Profile

Roll Number

2410030981

Institute

IILM University, Greater Noida, U.P.

Programme

B.Tech CSE, Batch 2024–2028

Internship Domain

Google AI-ML

Duration

8 Weeks | April – June 2026

ABOUT THE PROGRAM

EduSkills, AICTE & Google for Developers

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills.” It partners with the All India Council for Technical Education (AICTE) to run the AICTE–EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Google AI-ML track, EduSkills collaborates with Google for Developers to deliver an industry-aligned curriculum through the Google Developer Program’s learning pathways, quizzes, and skill badges.

ISO Certified

ISO 9001:2015 & ISO/IEC 20000-1:2018 certified organization.

AICTE Patronage

Run under the AICTE National Internship Portal.

Google for Developers

Industry curriculum delivered via the Google Developer Program.

PROBLEM STATEMENT

Bridging Theory and Applied Computer Vision

Although many engineering students learn the theoretical foundations of machine learning and neural networks in coursework, very few get structured, hands-on exposure to building and applying computer-vision models using an industry-standard framework such as TensorFlow. Concepts such as convolutional neural networks, on-device object detection, and image classification are often understood only theoretically — with little practice converting and integrating such models into real applications.

This internship addressed that gap — practically learning the full TensorFlow-based workflow:

Build & Train → **Detect Objects** → **Search Products** → **Integrate to App**

PROJECT OBJECTIVES

What the Internship Set Out to Achieve

Build CNNs with TensorFlow

Understand ML from first principles and build convolutional neural networks for image recognition.

Object Detection

Integrate pretrained detectors and train custom models with TensorFlow Lite & Model Maker.

Product Image Search

Build a visual, on-device product search feature across static images and live camera feeds.

Backend Integration

Build backend integration for a product image search mobile application.

Image Classification

Build and improve custom image-classification models.

App Integration

Create and integrate a custom image classifier into a working application.

TECHNOLOGIES & TOOLS

The Toolkit Behind the Internship

Python
Core programming language

TensorFlow & Keras
Model building & training

TensorFlow Lite
On-device deployment

Model Maker
Custom model training

CNNs
Image recognition

Object Detection
Pretrained & custom detectors

Product Image Search
Visual search integration

Image Classification
Custom classifiers

SYSTEM ARCHITECTURE

From Trained Model to Mobile Application

Model Layer	CNN built & trained in TensorFlow
Conversion Layer	Converted via TensorFlow Lite & Model Maker
Detection Layer	Pretrained & custom object detectors
Search Layer	Product image search, backend integrated
Classification Layer	Custom image classifier refines results
Application Layer	Integrated into a mobile application

METHODOLOGY

The 8-Week Learning Journey

1 WEEK

Neural Networks with TensorFlow

CNNs for image recognition & classification

2 WEEK

Get Started: Object Detection

Integrate pretrained detectors into mobile apps

3 WEEK

Go Further: Object Detection

Custom models with TFLite & Model Maker

4 WEEK

Get Started: Product Image Search

On-device object detection for product search

5 WEEK

Product Image Search (contd.)

Static images & live camera feed detection

6 WEEK

Go Further: Product Image Search

Backend integration for mobile search app

7 WEEK

Go Further: Image Classification

Custom classification models

8 WEEK

Go Further: Image Classification

Integrate custom classifier into an app

EXPECTED OUTCOMES

Skills & Capabilities Gained

- Ability to explain and apply first-principles ML to build CNNs using TensorFlow
- Practical understanding of object detection with pretrained & custom TFLite models
- Hands-on exposure to building a product image search feature with backend integration
- Capability to build, evaluate and improve custom image-classification models
- Skill to integrate trained, converted TensorFlow Lite models into a mobile application
- Completed all learning-pathway modules & badges; cleared the Final Credential Validation with an “Excellent” grade

CERTIFICATION

Successfully Completed

Grade

E – Excellent

Certificate ID

479384cd2c13d79088f5

Student ID

STU6997eaa57f1f91771563685

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Thank You

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